

A Study of 3D Character Control Methods_Keyboard, Speech, Hand Gesture, and Mixed Interfaces

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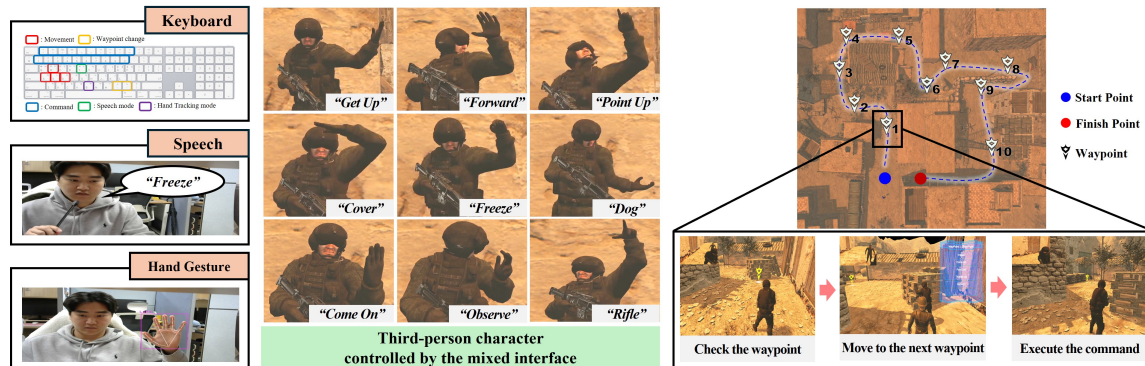


Figure 1: A user controls a character in a third person view point (e.g., behind the back) through the desktop interfaces, namely, by keyboard input, speech command, and hand gestures. The task involved navigating the character to waypoints in the virtual space and executing commands to activate the character's behavior/animation using the mixed interfaces.

Abstract

One popular form of a tele-presence guidance VR/MR system features local trainees utilizing immersive interfaces (such as headset and controllers), while the remote trainer employs the desktop interface, for the sake of familiarity, efficiency and convenience, to control the presence and actions of one's avatar in the third person perspective. Nowadays, the desktop (or non-VR) interfaces may extend from the mere keyboard/mouse to include speech and upper body/arm/hand gestures as well. Given this, however, it is not clear whether there exist any guideline in terms of what kind of interfaces might offer the best usability for effective avatar control. This poster presents a pilot study investigating such a problem with regards to using the keyboard, speech, and hand gestures (and properly mixing them) for a MR-based military training environment where the user controlled a squad leader character for making tactical signals to following squad members. Results indicated that appropriately mixing the interfaces by the characteristics of the subtask produced the highest usability and preference, reduced world load, and even the sense of embodiment with their avatar.

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1 introduction

In a tele-presence “guidance” VR/MR system, it may not be necessary to provide immersion/presence to the expert trainer as long as the instructions can be conveyed, which in many cases may lead to adopting the popular and familiar desktop interface and using the third-person viewpoint. In a related context, controlling the actions of one's avatar in such a way is quite prevalent in 3D games.

However, for situations where complex control of the avatar is required (e.g., in military tactical signal training), the keyboard/mouse may not suffice because of the sheer number of commands for various poses and their combinations. Note that the desktop interfaces may extend from the keyboard/mouse to speech and body/arm/hand gestures. However, even using speech commands or acting them out with gestures may be problematic for other reasons. For instance, certain commands need to be delivered silently, navigation commands are not appropriate with as-is gestures, or simple often-used commands may be better off and less tiring with direct hot-keys.

It seems clear that the most effective interface for avatar control would require a design that combines the desktop interfaces' strengths; however, no clear guideline exists. This poster introduces a mixed-interface design for controlling a 3D character in the third-person viewpoint and presents pilot user evaluation results.

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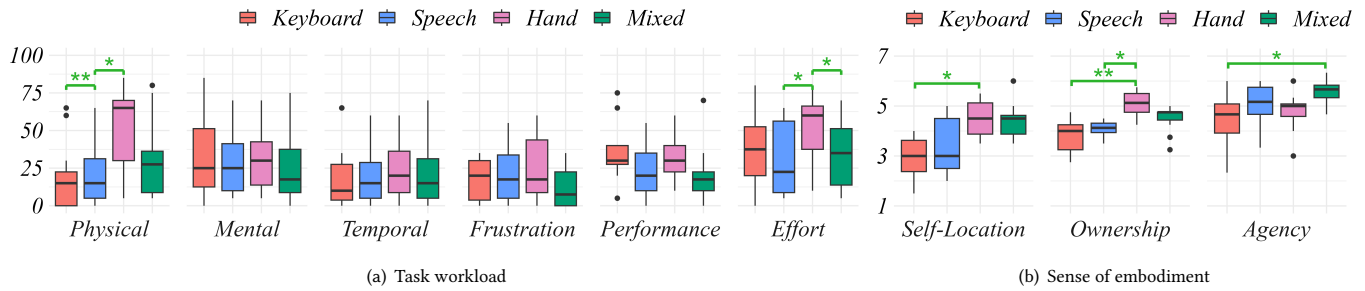


Figure 2: The avatar control task workload and the sense of embodiment (* $p < .05$ and ** $p < .01$).

2 Experimental Set-up

The interface designs were evaluated and compared through an experimental military simulation environment (implemented using Unity) where the trainer controlled a squad leader character for making tactical signals to following squad members. Twenty distinct signals from references such as the U.S. Army's Visual Signals (TC 3-21.60) were available (see Figure 1). As for the experimental task, the user was asked to navigate the leader character to ten waypoints strategically placed at regular intervals (see Figure 1 (right)); then, once at the waypoint, to execute, invoke and make commands for 5 or 6 tactical signals, such as "forward," "freeze," and "cover." Over the course of the whole task, the user had to make a total of 54 tactical commands, some of which were repeated.

Four interfaces, for controlling the 3D avatar to make the hand signals, were compared: (1) keyboard, (2) speech, (3) hand gestures, and (4) a mixed interface (see Figure 1 (left)). Note that for all cases, the keyboard was used for navigational control (WASD keys) and the mouse for viewpoint control, as a way to adopt the most prevalent and basic form of avatar control desktop interface.

The *keyboard* was designed for simplicity, with 20 tactical commands being mapped to each of the number keys from 1 to 0, and to the function keys from F1 to F10. The *speech* interface employed a speech recognition API and utilized a predefined rule map that linked words to animations. For *hand* gestures, the hand tracking API was employed to obtain the user's hand poses from the camera, and the tracked data was used to drive a retargeted character animation. Lastly, in *mixed interface*, users were allowed to selectively use any of the three interfaces according to their preferences.

3 Experiment and Results

The experiment was run as a 1×4 within-subject. Evaluation metrics included subjective reports on task workload (NASA-TLX), system usability (SUS), and character embodiment (self-location, ownership, and agency) [Gorisse et al. 2017]. 12 participants (7 males, aged 21 to 35 ($M = 26.2$; $SD = 4.7$)) were recruited.

Statistical analysis used the Friedman test due to the small sample size. For pairwise comparisons, the Wilcoxon signed-rank test with Bonferroni correction was applied, with a significance level of 5%.

System usability results were as follows: *keyboard* (63.3 ± 15.1), *speech* (68.3 ± 17.3), *hand* gestures (54.1 ± 12.6), and *mixed interface* (66.6 ± 8.1). Despite the mixed interface ranking second, there was no significant difference among conditions ($\chi^2 = 7.48$; $p = .058$).

Task workload results, as illustrated in Figure 2 (a), showed significant differences in physical demand ($\chi^2 = 18.92$; $p < .001$) and

effort ($\chi^2 = 12.49$; $p = .006$). Specifically, *hand* gesture required more physical effort than *keyboard* ($p = .002^{**}$) and *speech* ($p = .014^{*}$), and nearly so than *mixed interface* ($p = .071$). *Hand* gestures also demanded more effort than *speech* ($p = .035^{*}$) and *mixed interface* ($p = .023^{*}$). This increased demand was due to the challenge of maintaining raised hands and making precise gestures.

For the embodiment, reflecting the perceived sense of merging an avatar, significant differences were found in all factors (see Figure 2 (b)). For self-location ($\chi^2 = 13.2$; $p = .004$), the *hand* gesture was significantly higher than in the *keyboard* ($p = .032^{*}$). The *mixed interface* outperformed *keyboard* but did not reach significance ($p = .086$). Ownership showed significant differences ($\chi^2 = 20.3$; $p < .001$), with *hand* gestures scoring higher than *keyboard* ($p = .001^{**}$) and *speech* ($p = .011^{*}$). In particular, the factor of Agency, which measures the user's sense of control and influence over their character ($\chi^2 = 10.1$; $p = .017$), showed that the *mixed interface* was significantly higher than *keyboard* ($p = .035^{*}$).

4 Discussion and Conclusion

In summary, participants felt a higher sense of embodiment when using hand gestures over the keyboard/speech. This likely stems from the direct translation of their movements into the character. However, hand gestures increased the task workload. Thus, a mixed interface could serve as an effective alternative, as some commands are executed using the keyboard/speech while others are performed through hand gestures. This strategy reduced the task load and enhanced the user experience, showing significant potential for improving the sense of control and agency over their characters.

This study indicated that this mixed interface has the potential to preserve adequate usability, not significantly increase task workload, and enhance the sense of embodiment. Our findings indicate that sticking to a single mode for consistency may not always be the best choice, and the individual subtask characteristics should be considered when choosing the interface.

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